

Learned variable-support priors accelerate symbolic regression of physical laws

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Abstract

Symbolic regression recovers closed-form, interpretable physical laws from data, but its cost grows steeply with the number of candidate variables and operators, most of which never appear in the governing equation. Here we show that a learned, observation-conditioned front-end that predicts which variables are relevant—a support prior—substantially improves and accelerates symbolic regression without changing the backend solver. The front-end couples a graph-attention network over a variable co-occurrence graph with a statistical-feature model, and recovers the true variables with recall 1.000 across three physics suites, so it essentially never discards a real variable. Supplied to the evolutionary backend PySR under a fixed time budget, the prior doubles exact-law recovery (from 30% to 60%), reduces the search space by 86%, and accelerates time-to-solution by 3–6×; on black-box tabular data it improves accuracy while a same-size random-support control degrades it, isolating learned guidance from dimensionality reduction. The approach is solver-agnostic, adds ~ 40 ms per task, and keeps final law selection with a verifiable symbolic backend.

1 Introduction

The automated discovery of governing equations from observation is a long-standing goal of the physical sciences^{1–3}. Symbolic regression (SR) addresses it directly: rather than fitting an opaque function, SR searches the space of mathematical expressions for a compact formula that both fits the data and remains human-interpretable^{4,5}. The resulting equation can be dimensionally checked, extrapolated and falsified, properties that black-box predictors lack, which has made SR central to data-driven physics, from rediscovering conservation laws to proposing constitutive relations^{2,6,7}.

The central obstacle is combinatorial. An SR backend searches over expression trees whose leaves are drawn from the input variables and whose internal nodes are drawn from an operator library; the number of candidate trees grows exponentially with the number of admissible leaves and operators and with tree depth^{8,9}. In physics this cost is largely wasted: a governing law typically uses only a handful of the measured variables and a small subset of the operator library, yet a solver run over all columns and operators still explores an enormous number of branches that cannot appear in the answer. Established systems attack this from the solver side with dimensional analysis, decomposition, sparsity priors or efficient enumeration^{2,6,10,11}. A separate line learns to decode the equation directly from numerical data with transformer or reinforcement-learning models^{12–15}, sometimes coupling a neural proposer with explicit search^{16,17} or multimodal pre-training^{18,19}, with e-graph, diffusion and generative-latent variants^{20–22}. These end-to-end models advance quickly but remain unreliable at *assignment*: they often recover the right ingredients yet place them in the wrong terms, so exact symbolic equivalence stays low.

We take a complementary route that sidesteps this brittleness. Instead of designing a faster search or decoding the whole equation, we *narrow the space the search must explore* (Fig. 1). Given a small observation set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^q$ with q as small as 50–100, a learned model predicts which input columns (and, as an auxiliary, which operators) are likely to participate in the law, and passes this *support prior* to a conventional SR backend that performs the final symbolic search and model selection. Predicting support is a far easier and more robust target than decoding the full expression, and a perfect-recall variable filter hands the backend an exponential head start while never making the true law unreachable.

In this work we cast variable- and operator-support prediction as a reusable, backend-agnostic SR front-end; we present a graph-attention support predictor, ensembled with a statistical-feature model, that recovers the true variables with recall 1.000 across three physics suites; and we show that supplying this prior to a fixed-budget backend doubles exact-law recovery, reduces the search space by 86% and accelerates time-to-solution by 3–6 $\times$, while a same-size random-support control confirms that the benefit is learned search guidance rather than dimensionality reduction.

2 Results

The front-end recovers every true physical variable

We first evaluate the support prior intrinsically, without any backend. For each task the front-end receives $q = 100$ noisy observations whose columns include the genuine variables together with random “distractor” columns, and must predict the true variable subset. Across three suites of physical laws—AI-Feynman (118 equations), a curated real-law benchmark (real591, 91 tasks) and an extended physics-law set (1085 equations)—the predictor attains recall 1.000 (or 0.999) with perfect recall on essentially every task (Fig. 2). It almost never drops a variable that the law uses; its only error mode is occasionally retaining a few extra noise columns, which merely costs the backend a little additional search rather than making the law unreachable. Precision ranges from 0.85 to 1.00, reflecting an intentional high-recall operating point: removing a true variable is fatal to recovery, whereas keeping a spurious one is recoverable downstream. A naive statistical baseline trained on few records with name-matched labels reaches only recall 0.48, confirming that the graph-attention model contributes the decisive signal.

A perfect-recall prior strengthens a fixed-budget solver

We next insert the front-end before PySR⁵ and compare three conditions at a fixed 10 s per-task budget on a 30-equation Feynman subset with ~ 20 distractor columns: (A) full PySR over all columns and the full 17-operator library; (B) +variable prior, restricting the columns to the predicted support; and (C) +operator prior, additionally restricting the operator library. Restricting variables raises exact-law recovery (held-out $R^2 \geq 0.999$) from 9/30 (30%) to 18/30 (60%), lifts mean held-out R^2 from 0.919 to 0.967, reduces the average column count from 23.2 to 3.2 (an 86% reduction) and halves the effective search vocabulary from 40 to 20 (Fig. 3). The effect is consistent across individual formulas (Fig. 4): the variable prior moves nearly every formula toward higher held-out R^2 and converts many budget-limited failures of full search into exact recoveries, by removing branches that could never appear in the law.

The gain is search guidance, not dimensionality reduction

A natural concern is that any reduction in column count would help. We control for this with a same-size random-support baseline that retains the same number of columns but selects them without using the observations. On 24 PMLB/SRBench black-box tabular tasks (three seeds, $q = 100$, 5 s budget), the learned support raises mean held-out R^2 from 0.347 to 0.495 with 10 distractor variables and from 0.315 to 0.488 with 20 distractors, using only 31% and 18% of the input columns; the same-size random support instead degrades performance to 0.220 and -0.028 (Fig. 5). At matched column count the random control is far worse than full search whereas the learned prior is better—direct evidence that the front-end supplies observation-conditioned information about *which* columns matter, not merely *how many*, and that it helps even when no symbolic ground truth exists.

Pruning the search space accelerates time-to-solution

Because variable filtering shrinks the terminal set of the expression tree from ~ 24 to ~ 4 symbols—an exponential reduction in branching—it also shortens the time to reach an exact fit. Measuring time-to-solution with early stopping on exact recovery, the front-end yields $3\text{--}6\times$ speedups on typical physics formulas (Fig. 6); in some cases it converts a within-budget failure of full search into a fast success. The exception is deep-tree formulas whose bottleneck is expression depth rather than variable count, where pruning the variable set gives little benefit. Front-end inference adds only ~ 40 ms per task, near-constant in the number of observations (Fig. 7) and negligible against the seconds of backend search it saves.

Operator priors are net-neutral but a useful safeguard

Restricting the operator library in addition to variables (condition C) does not improve mean exact recovery (it is slightly lower, 14/30), because operators are intrinsically harder to infer from small, noisy domains and a mis-removed operator is fatal. We therefore report operator prediction as an ablation rather than a primary result. It nonetheless has a clear conditional value: when full-operator search returns a pathological high- R^2 fit built from exotic operators, restricting the library to the predicted operators reliably recovers a sane solution (Fig. 8; e.g. held-out R^2 improving from 0.19 to 0.92). The triggering signatures—an exotic low-probability operator, anomalous complexity, or R^2 stuck just below 1—are visible at inference without the ground truth, so an adaptive “re-run with operator restriction when the fit looks pathological” policy is deployable in practice.

3 Discussion

These results support a simple thesis: for symbolic regression of physical laws, a learned support prior is a high-leverage, low-risk intervention. Because a perfect-recall variable filter never makes the true law unreachable, it can be applied unconditionally; because it removes an exponential number of irrelevant branches, it materially improves both the success rate and the speed of a fixed-budget backend. The front-end is solver-agnostic—its output is a reduced variable/operator set that any tree-search, enumeration or generative backend^{5,11,22} can consume—and cheap enough to prepend to every run. The same-size random-support control shows the benefit is genuine search guidance rather than dimensionality reduction, complementing the dimensional and physics-aware priors that already strengthen scientific SR^{2,7}.

Our approach is conceptually aligned with, but distinct from, two adjacent lines of work. Physics-aware neural models embed structure such as Hamiltonian or Lagrangian form, differential-equation

structure, or symbolic distillation of trained networks^{23–26}, and concept-driven or abductive discovery systems^{27–29} likewise aim to keep the final law symbolic and verifiable; we contribute the upstream step of deciding *which measured quantities enter the law at all*. Because the front-end consumes *sets* of observations, it builds on permutation-invariant and set-attention architectures^{30,31}, and its representation could be sharpened with the contrastive and geometric objectives used to prevent representational collapse or encode hierarchy^{32–34}.

The honest limitation, shared with direct neural SR^{13,14}, is that the neural component cannot yet reliably *assign* variables and operators to the correct positions in the expression; this is precisely why we leave final term assignment to a symbolic, verifiable backend rather than emitting the equation end-to-end. Closing this assignment gap—through richer compositional latents, explicit constant and exponent recovery, and objectives that keep the observation and formula representations aligned^{18,35}—is, in our view, the central open problem in moving from a front-end prior toward reliable end-to-end neural discovery^{36–38}. Even short of that goal, the present front-end is immediately useful: it makes existing physical-law solvers both more successful and several times faster, with a provably safe failure mode.

4 Methods

Problem setup. Each task provides $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^q$ with $\mathbf{x}_i \in \mathbb{R}^m$, where only an unknown subset $S^* \subseteq \{1, \dots, m\}$ of columns appears in the target law f^* . The front-end predicts $\hat{S} \approx S^*$ (and an optional operator set $\hat{O}$) and passes it to an SR backend run under a fixed time budget. Tasks are padded with random distractor columns to emulate the realistic regime in which a measurement table contains many irrelevant channels.

Front-end architecture. The variable-support predictor is an ensemble of two complementary models. (1) A graph-attention network^{30,31} operates on a graph whose nodes are the input columns; node features combine per-column statistics (moments, correlation with the target, monotonicity and dimensional descriptors) and edges encode a variable co-occurrence prior learned from a knowledge graph of physical laws together with observation-derived correlation. Attention over this graph lets the model reason about which columns *jointly* support a plausible law rather than scoring columns in isolation. (2) A random forest scores each column from the same per-column features. The final score is $s(\text{col}) = \text{RF}(\text{col}) + \sigma(\text{GAT}(\text{col}))$, and a column is retained if $s \geq \tau$ with a low threshold ($\tau = 0.10$) chosen to favour recall. An analogous graph-attention model with graph-level pooling produces the multi-label operator prediction used for the optional operator prior.

Training. The predictors are trained only on internal physical-law records—formula templates with sampled observation tables drawn from a curated knowledge graph of physical laws—never on the external evaluation suites. Training pools of 8000–40 000 formulas were used, with observation tables pre-sampled across variable ranges and a 10–30 distractor curriculum so the front-end is robust to the number of irrelevant columns at test time. An adversarial variant (a generative-adversarial graph-attention model) was explored but gave no consistent gain and is not used.

Backend integration and evaluation. We use PySR⁵ as the downstream backend because it exposes variable and operator restrictions directly and is reproducible locally; the same prior interface applies to enumeration¹¹ and generative²² backends. All comparisons fix the backend budget across conditions, so the only changed quantity is the search space exposed to the solver. Exact recovery denotes held-out $R^2 \geq 0.999$. Support quality is measured by precision and recall against the known law variables; the same-size random-support control retains the same number of columns selected without the front-end. Time-to-solution is measured with early stopping on exact recovery, and front-end overhead is measured separately. All reported numbers are produced by the

released evaluation scripts from the recorded result artifacts.

Data availability

The external benchmarks (AI-Feynman, PMLB/SRBench) are public^{2,8,39}; the physical-law training records are materialised from public formula and range information. The recorded result artifacts that back every figure are released with the code.

Code availability

Code for the front-end, training, evaluation and figure generation, together with the result artifacts and the trained graph-attention weights, is released at <https://github.com/<anonymized>> under the MIT License.

Author contributions

The author(s) conceived the study, designed and implemented the front-end and evaluation, performed the experiments and analysis, and wrote the manuscript. (Detailed per-author contributions will be added after de-anonymisation.)

Competing interests

The authors declare no competing interests.

References

- [1] Schmidt, M. & Lipson, H. Distilling free-form natural laws from experimental data. *Science* **324**, 81–85 (2009).
- [2] Udrescu, S.-M. & Tegmark, M. Ai feynman: A physics-inspired method for symbolic regression. *Science Advances* **6**, eaay2631 (2020).
- [3] Makke, N. & Chawla, S. Interpretable scientific discovery with symbolic regression: a review. *Artificial Intelligence Review* **57**, 2 (2024).
- [4] Koza, J. R. *Genetic Programming: On the Programming of Computers by Means of Natural Selection* (MIT Press, 1992).
- [5] Cranmer, M. Interpretable machine learning for science with pysr and symbolicregression.jl. *arXiv preprint arXiv:2305.01582* (2023).
- [6] Brunton, S. L., Proctor, J. L. & Kutz, J. N. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences* **113**, 3932–3937 (2016).
- [7] Tenachi, W., Ibata, R. & Diakogiannis, F. I. Deep symbolic regression for physics guided by units constraints: toward the automated discovery of physical laws. *The Astrophysical Journal* (2023).
- [8] La Cava, W. *et al.* Srbench: A living benchmark for symbolic regression. *IEEE Transactions on Evolutionary Computation* **28**, 1276–1290 (2024).

- [9] Petersen, B. K. *et al.* Deep symbolic regression: Recovering mathematical expressions from data via risk-seeking policy gradients. In *International Conference on Learning Representations* (2021).
- [10] Rudy, S. H., Brunton, S. L., Proctor, J. L. & Kutz, J. N. Data-driven discovery of partial differential equations. *Science Advances* **3**, e1602614 (2017).
- [11] Ruan, K. *et al.* Discovering physical laws with parallel symbolic enumeration. *Nature Computational Science* (2025).
- [12] Biggio, L., Bendinelli, T., Neitz, A., Lucchi, A. & Parascandolo, G. Neural symbolic regression that scales. In *International Conference on Machine Learning* (2021).
- [13] Kamienny, P.-A., d’Ascoli, S., Lample, G. & Charton, F. End-to-end symbolic regression with transformers. *Advances in Neural Information Processing Systems* **35**, 10269–10281 (2022).
- [14] Landajuela, M. *et al.* A unified framework for deep symbolic regression. *Advances in Neural Information Processing Systems* **35**, 33985–33998 (2022).
- [15] d’Ascoli, S., Kamienny, P.-A., Lample, G. & Charton, F. Deep symbolic regression for recurrence prediction. In *International Conference on Machine Learning*, 4520–4536 (2022).
- [16] Shojaee, P., Meidani, K., Barati Farimani, A. & Reddy, C. K. Transformer-based planning for symbolic regression. *arXiv preprint* (2024).
- [17] Kamienny, P.-A., Lample, G., Lamprier, S. & Virgolin, M. Deep generative symbolic regression with Monte-Carlo tree search. In *International Conference on Machine Learning* (2023).
- [18] Meidani, K., Shojaee, P., Reddy, C. K. & Barati Farimani, A. SNIP: Bridging mathematical symbolic and numeric realms with unified pre-training. In *International Conference on Learning Representations* (2024).
- [19] Li, Y. *et al.* MMSR: Symbolic regression is a multi-modal information fusion task. *arXiv preprint arXiv:2402.18603* (2024).
- [20] Zheng, H., Wang, S., Gangwar, N. & Kani, N. E-gen: Leveraging e-graphs to improve continuous representations of symbolic expressions. *arXiv preprint* (2025).
- [21] Bastiani, Z., Kirby, R. M., Hochhalter, J. & Zhe, S. Diffusion-based symbolic regression. *arXiv preprint arXiv:2505.24776* (2025).
- [22] Li, Q., Hu, Y., Liu, J. & Chen, Y. GenSR: Symbolic regression based on equation generative space. In *The Fourteenth International Conference on Learning Representations* (2026). URL <https://openreview.net/forum?id=8emIjwUQZg>.
- [23] Greydanus, S., Dzamba, M. & Yosinski, J. Hamiltonian neural networks. *Advances in Neural Information Processing Systems* **32** (2019).
- [24] Cranmer, M. *et al.* Lagrangian neural networks. *arXiv preprint arXiv:2003.04630* (2020).
- [25] Chen, R. T. Q., Rubanova, Y., Bettencourt, J. & Duvenaud, D. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems* (2018).
- [26] Cranmer, M. *et al.* Discovering symbolic models from deep learning with inductive biases. In *Advances in Neural Information Processing Systems* (2020).
- [27] Fang, Y.-L., Jian, D.-S., Li, X. & Ma, Y.-Q. Ai-newton: A concept-driven physical law discovery system without prior physical knowledge. *arXiv preprint* (2025).
- [28] Srivastava, K., Dash, S., Cory-Wright, R., Trager, B. & Cornelio, C. Bridging the gap between scientific laws derived by ai systems and canonical knowledge via abductive inference with ai-noether. *arXiv preprint* (2025).
- [29] Ying, J. *et al.* A neural symbolic model for space physics. *arXiv preprint* (2025).
- [30] Zaheer, M. *et al.* Deep sets. In *Advances in Neural Information Processing Systems* (2017).
- [31] Lee, J. *et al.* Set transformer: A framework for attention-based permutation-invariant neural

- networks. In *International Conference on Machine Learning* (2019).
- [32] Bardes, A., Ponce, J. & LeCun, Y. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations* (2022).
 - [33] Nickel, M. & Kiela, D. Poincare embeddings for learning hierarchical representations. In *Advances in Neural Information Processing Systems* (2017).
 - [34] Sinha, A., Zeng, S., Yamada, M. & Zhao, H. Learning structured representations with hyperbolic embeddings. In *Advances in Neural Information Processing Systems* (2024).
 - [35] Rogers, B. *et al.* What we don’t c: Manifold disentanglement for structured discovery. *arXiv preprint* (2025).
 - [36] Voigt, H., Kahlmeyer, P., Lawonn, K., Habeck, M. & Giesen, J. Analyzing generalization in pre-trained symbolic regression. *arXiv preprint* (2024).
 - [37] Shojaee, P., Meidani, K., Gupta, S., Barati Farimani, A. & Reddy, C. K. LLM-SR: Scientific equation discovery via programming with large language models. In *International Conference on Learning Representations* (2025).
 - [38] Hallin, A. *et al.* Universal new physics latent space. *arXiv preprint* (2024).
 - [39] Olson, R. S., La Cava, W., Orzechowski, P., Urbanowicz, R. J. & Moore, J. H. PMLB: a large benchmark suite for machine learning evaluation and comparison. *BioData Mining* **10** (2017).

Figures

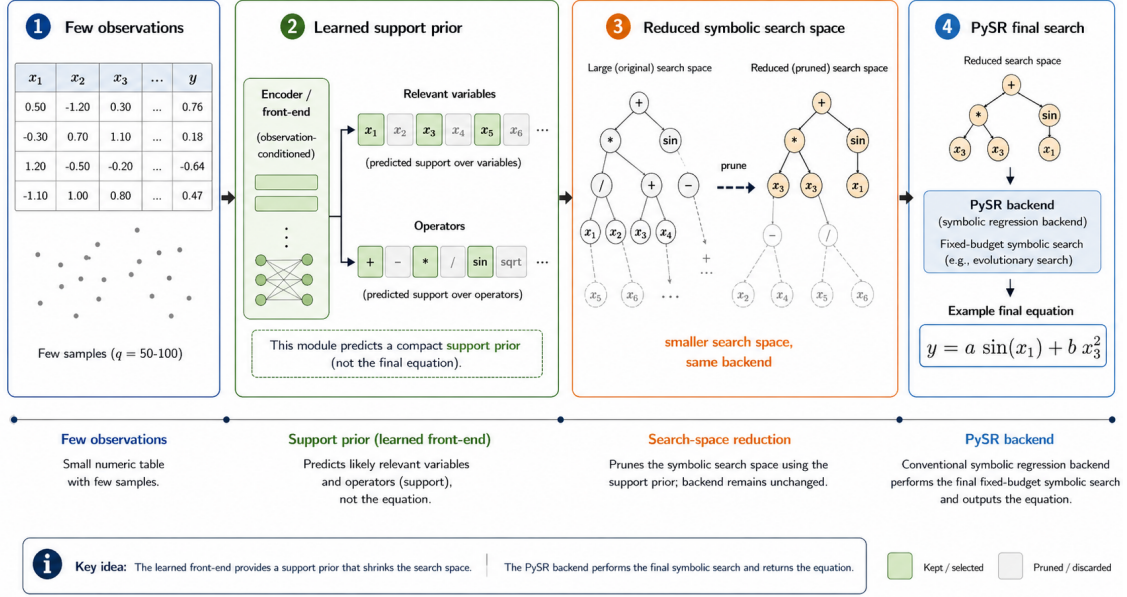


Fig. 1. A learned front-end narrows the symbolic-regression search space. From a small set of observations the front-end predicts a compact support prior (relevant variables and, optionally, operators); a conventional symbolic-regression backend then performs the final search over the reduced space and returns a verifiable equation. The neural model proposes where to look; the backend decides what the law is.

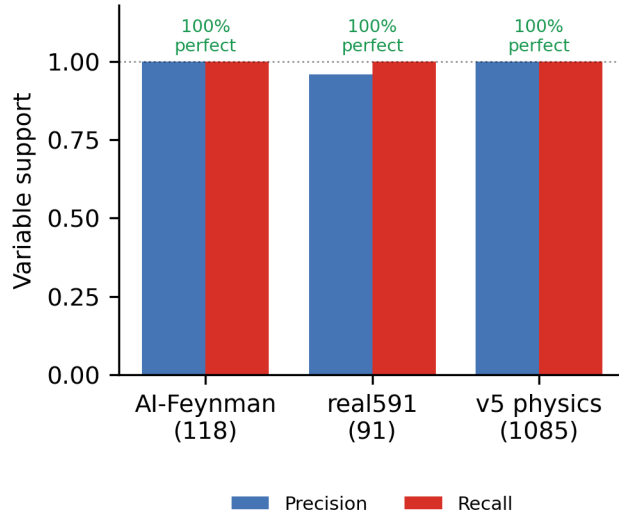


Fig. 2. The front-end recovers every true variable. Precision and recall of predicted variable support across three physics suites. Recall is 1.000 (AI-Feynman, real591) or 0.999 (extended set), with perfect recall on $\sim 100\%$ of tasks; the high-recall operating point trades a little precision for a safe failure mode in which a true variable is essentially never dropped.

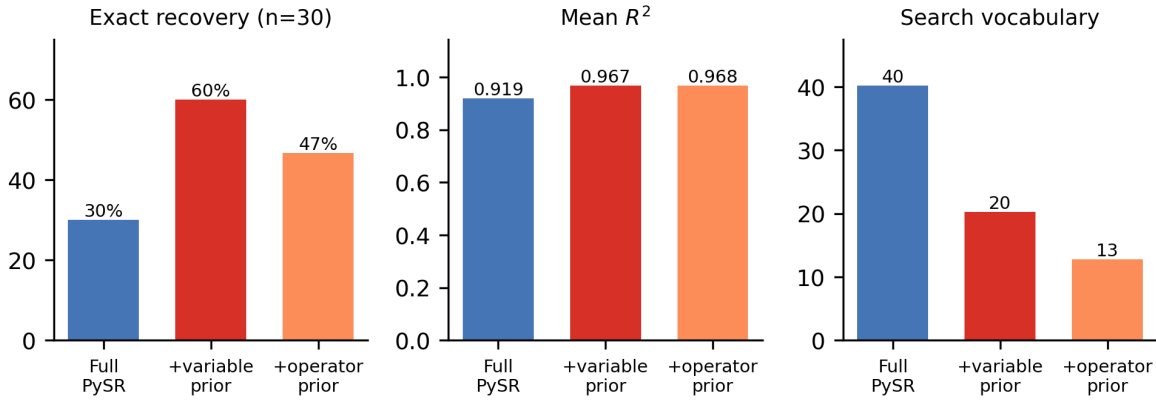


Fig. 3. A variable prior strengthens a fixed-budget solver. Full PySR versus +variable prior versus +operator prior on a 30-equation Feynman subset (10 s budget, ~ 20 distractors). The variable prior doubles exact recovery (30% to 60%), raises mean held-out R^2 (0.919 to 0.967) and halves the search vocabulary (40 to 20); adding the operator prior cuts the vocabulary further but does not improve exact recovery.

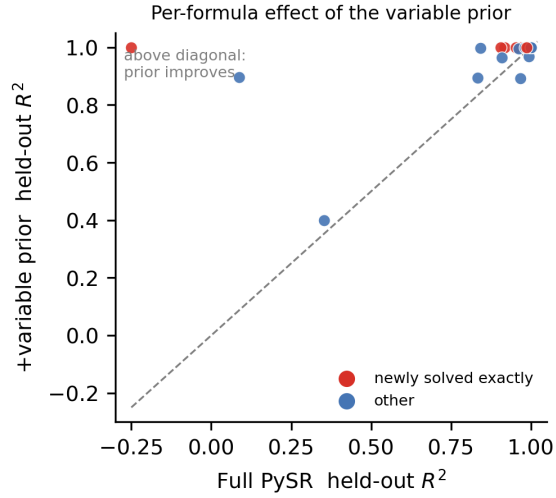


Fig. 4. The variable prior helps almost every formula. Held-out R^2 of each Feynman formula under full PySR (horizontal axis) versus the same backend with the +variable prior (vertical axis); points above the diagonal are improved. Highlighted points are formulas the prior newly recovers exactly that full search fails to recover within budget. Values are clipped at -0.25 for display.

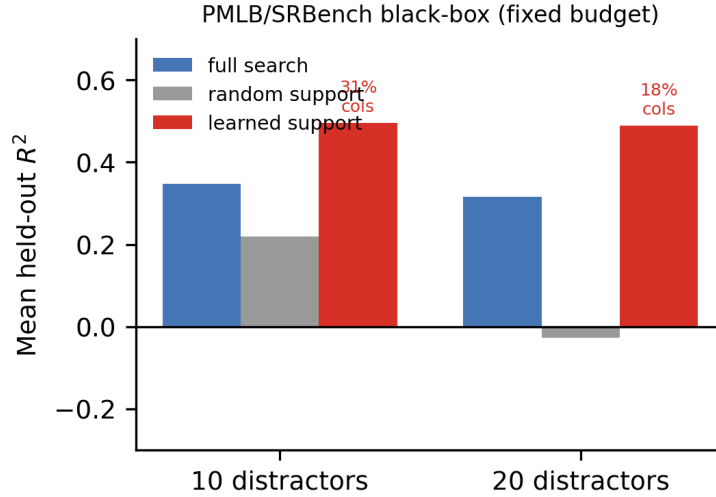


Fig. 5. The gain is search guidance, not fewer columns. Mean held-out R^2 on 24 PMLB/SRBench black-box tasks (three seeds, fixed 5 s budget) at 10 and 20 distractor variables. At matched, reduced column count (labels show columns relative to full search) the learned support improves over full search while a same-size random support degrades far below it.

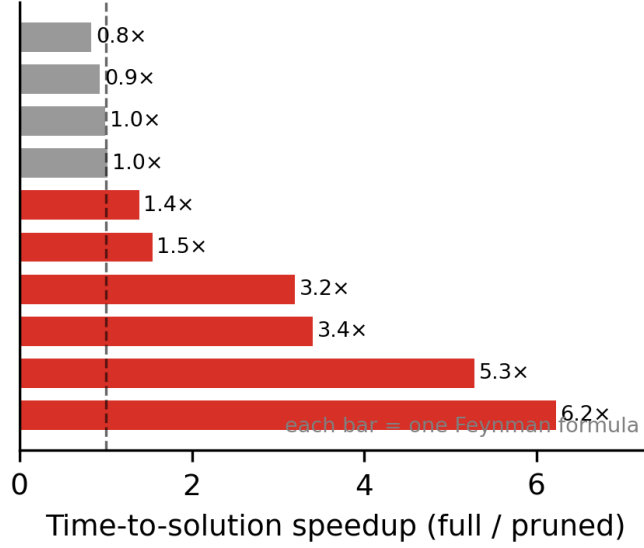


Fig. 6. Pruning the search space accelerates time-to-solution. Per-formula speedup (full-search time divided by pruned-search time) with early stopping on exact recovery. Typical physics formulas solve 3–6 $\times$ faster; deep-tree formulas (depth-, not variable-limited) show little gain.

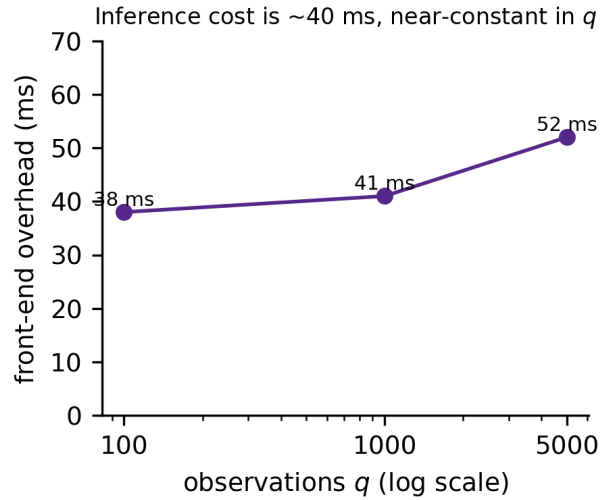


Fig. 7. The front-end is cheap. Front-end inference time (graph construction, random forest and graph-attention model) as a function of the number of observations q . The cost is ~ 40 ms and near-constant in q , negligible against the seconds of backend search it saves.

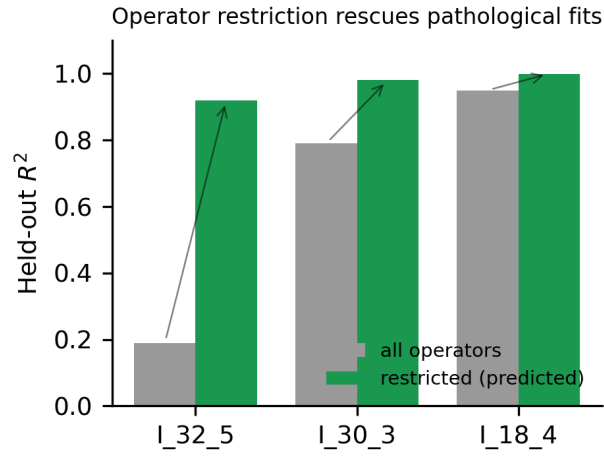


Fig. 8. Operator restriction rescues pathological fits. For cases where full-operator search returns a spurious high- R^2 solution built from exotic operators, restricting the library to the predicted operators recovers a sane fit (held-out R^2 before versus after restriction). Operator priors are net-neutral on average but valuable as an adaptive safeguard.